

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

- a) What are the disadvantages of Hill Climbing Search Technique? 3
Mention the possible solution(s) for each problem.
- b) Explain the working of Perceptron Network. 3
- c) Write a compete prolog program for reversing elements in a list. 3
- d) Explain the principles of means_end_analysis approach to problem solving. 2

Q-2

- [A] Write rule in prolog *getn (list, n, p)* where *p* returns the *n*th element from list. 4
- [B] A problem solving search can proceed either in forward or backward. 4
Whether search should proceed forward or backward for the following problems. Justify the answer with the reason.
1. Planning a party.
 2. Going back to home from unknown place.
- [C] Will the following literals Unify? If yes, then write the most general unifier. 4
 $W = \{ P(a, x, f(g(y))), P(z, f(z), f(u)) \}$

OR

Q-2

- [A] What are the basic knowledge representation issues? What are the 4
considerations in knowledge representation?
- [B] Write down the **water jug problem**. Give the rules for **water jug problem**. 4
- [C] Describe in brief various steps involved in NLP. 4

Q-3

- [A] Justify the use of *fuzzy logic* in AI. What are the criticisms for *fuzzy logic*? 4
- [B] Comparison of *Expert System* with *Traditional Software System*. 4
- [C] Draw the parse tree and write down the grammar rules for: 4
The boy smoked a cigarette.

OR

Q-3

- [A] Distinguish between *state space search* and *constraint satisfaction* technique. 4
What are the termination conditions for *constraint satisfaction* technique?
- [B] Write a prolog program to read *N* elements and store into the list. 4
- [C] Explain *Horn clauses* and execution strategy of PROLOG program. 4

SECTION-II

Q-4

- a) How are the *frames* organized? What are the advantages and disadvantages of *frames* and *semantic nets*? Why are the *scripts* required? Can it be considered as a variant of *frames*? 3
- b) Explain *cut* mechanism in PROLOG. Differentiate **red** and **green cut** in prolog. 4
- c) Explain the difference between *propositional* logic and *predicate logic* (FOPL). 4

Q-5

- [A] Explain the characteristic features of Expert System. 4
- [B] What is the main drawback of **back-propagation** network? What is the reason of it? Whether *momentum* term is required for learning in a back-propagation network? What is the effect of this term? 4
- [C] State and prove **Bay's Theorem**. Justify the use of **Bay's theorem** in **Bayesian Belief Networks**. 4

Q-5

OR

- [A] Differentiate: *Abduction* and *Inheritance*. 4
- [B] Differentiate: *Forward Reasoning* versus *Backward Reasoning*. 4
- [C] Which factors determine the choice of direction for a particular problem? 4

Q-6

- [A] **Transfer following into FOPL (First Order Predicate Logic):** 4
 1. All employees earning more than 60,000 Rs./year, pay income tax.
 2. No employee of University earns more than the VC.
 3. European vegetarians do not take milk, but Indian vegetarians do.
 4. Anything anyone eats and isn't killed by is fruit.
- [B] Write short note on: **Dempster-Shafer Theory**. 4

OR

- [B] Trace the execution of the constraint satisfaction procedure in solving the crypt arithmetic problem: **AND +OR= ALL**. 4
- [C] Explain: **Certainty Factors (CF)**. Justify the use of it in AI and Statistics. 4

OR

- [C] Where is the knowledge in **Neural Network** stored? Explain multilayer fully connected feed forward networks. 4